Prakhar Dixit

PhD Student in Computer Science | University of Maryland, Baltimore County

Baltimore, Maryland | prakhhar1997@gmail.com | +1 202 937 5716

Website | Google Scholar | LinkedIn | GitHub

I am a PhD student in Computer Science at the University of Maryland, Baltimore County. I study how AI systems reason, learn from experience, and use memory, from mathematical problem solving with language models to reinforcement learning and robotics.

Publications & Technical Writing

ISM: Self-Improving Strategy Memory for Continual Mathematical Reasoning

Prakhar Dixit and Tim Oates

ICML 2026 · AI4Math Workshop | Jun 2026

Paper | OpenReview | Code

Honest Lying: Understanding Memory Confabulation in Reflexive Agents

Prakhar Dixit, Sadia Kamal, and Tim Oates

ICML 2026 · Failure Modes in Agentic AI Workshop | Jun 2026

Paper

WildfireVLM: AI-powered Analysis for Early Wildfire Detection and Risk Assessment Using Satellite Imagery

Aydin Ayanzadeh, Prakhar Dixit, Sadia Kamal, and Milton Halem

IGARSS 2026 | Mar 2026

Paper

SOTA Results in Agentic Memory on LongMemEval

Prakhar Dixit et al. - Emergence AI

Emergence AI · Blog | Jul 2025

Blog Post

SBI-RAG: Enhancing Math Word Problem Solving for Students through Schema-Based Instruction and Retrieval Augmented Generation

Prakhar Dixit and Tim Oates

NeurIPS 2024 · MATH-AI Workshop | Nov 2024

Paper

Dynamic Edge Weighting in Relational Graph Convolutional Networks: Enhancing Sample Efficiency via Graph Attention in Reinforcement Learning

Prakhar Dixit - University of Maryland, Baltimore County

MS Thesis · ProQuest | Aug 2024

Thesis

ReProHRL: Towards Multi-Goal Navigation in the Real World using Hierarchical Agents

Tejaswini Manjunath, Mozhgan Navardi, Prakhar Dixit, Bharat Prakash, and Tinoosh Mohsenin

AAAI 2023 · RL Ready for Production Workshop | Jan 2023

Paper

Toward Real-World Implementation of Deep Reinforcement Learning for Vision-Based Autonomous Drone Navigation with Mission

Mozhgan Navardi, Prakhar Dixit, Tejaswini Manjunath, Nicholas R. Waytowich, Tinoosh Mohsenin, and Tim Oates

RSS 2022 · Sim2Real Workshop | Jun 2022

Paper

Research & Professional Experience

Applied Scientist Intern - Microsoft

Redmond, WA May 2026 - Present

- Working with the Microsoft Copilot Studio Team to improve the skills inside the MCS platform.

Summer Research Intern - Emergence AI

New York City May 2025 - Oct 2025

- Worked with a long-term memory system for conversational agents.
- Achieved SOTA on LongMemEval with 86% accuracy , surpassing the previous best by 15%.
- Developed specialised agents for enterprise tasks in planning and self-improvement.

Graduate Research Assistant - UMBC

Baltimore, MD Jan 2022 - Present

- Developed ISM , a self-improving strategy memory for continual mathematical reasoning (ICML 2026 · AI4Math Workshop).
- Enhancing mathematical reasoning in LLMs through human-driven learning techniques.
- Designed Sim2Real framework with YOLO object detectors - 85% transfer success to real-world robots.
- Enhanced nano drone navigation efficiency by 30% using Hierarchical RL.
- RLHF via TAMER (Scalar binary feedback) for sequential decision-making agents.
- Integrated graph attention networks in R-GCN - 20% improvement in sample efficiency.

Graduate Assistant - UMBC

Baltimore, MD Aug 2021 - Dec 2021

- Assisted in teaching CMSC 691.4 (Intro to Data Science) for 50+ students , ensuring 95% assignment completion.
- Collaborated with professor on student projects, increasing engagement rates by 20%.

Software Engineer - Titan Company Limited

Bengaluru, India Jul 2019 - Apr 2021

- Built Spring Boot + REST API web app for 8,000+ employees - 40% reduction in data retrieval time.
- Python dashboard for login analytics - 25% improvement in user engagement tracking.

Software Engineer Intern - Titan Company Limited

Bengaluru, India Mar 2019 - Jun 2019

- Improved software reliability by 30% , reduced bug occurrences by 20%.
- Tested BPM-related applications, increasing process efficiency by 25%.

Research Intern - National University of Singapore

Singapore Jun 2018 - Jul 2018

- Achieved 98.5% accuracy on credit card fraud detection - outperformed ANN, RF, SVM.
- Advanced preprocessing reduced data redundancy by 25% and training time by 30%.

Education

PhD · Computer Science | University of Maryland Baltimore County
Baltimore, MD · 2023 - Present | GPA 3.589
Research: LLM mathematical reasoning and improving sample efficiency in RL. Related work: ISM .

MS · Computer Science | University of Maryland Baltimore County
Baltimore, MD · 2021 - 2023 | GPA 3.5
Thesis: Dynamic Edge Weighting in R-GCN for Sample Efficiency in RL

B.Tech · CS & Engineering | SRM Institute of Science and Technology
Chennai · 2015 - 2019 | GPA 3.7
Final Year Project: Neural Sketching

Projects

ISM: Self-Improving Strategy Memory for Continual Mathematical Reasoning

A strategy memory that learns from successful and failed episodes to improve mathematical reasoning with a frozen language model. Evaluated on MATH-Hard and OlympiadBench.

Paper | OpenReview | Code

ScholarAgent

An AI agent that fetches recent ArXiv research papers based on user search - like Google Scholar but in a more user-friendly and readable format. Built with LLMs and agentic retrieval pipelines.

Code

LLM Detect AI Text

Developed a machine learning model to accurately detect whether an essay was written by a student or an LLM. Finished in the top 25% of the Kaggle competition leaderboard.

Code

Deep Learning Steering Model

Self-driving prototype using CNNs trained on dash-cam images to predict steering angles in real time. Built as part of the Team Camber Racing autonomous vehicle project.

GitHub

ISS_Info

Python package fetching real-time International Space Station data - location, crew, pass times and more. Over 60,000 downloads on PyPI.

GitHub

Lane Detection

Computer vision pipeline for road lane identification using Canny edge detection, Hough transforms, and perspective warping - part of the self-driving vehicle stack.

GitHub

Credit Card Fraud Detection

Applied logistic regression to a large imbalanced dataset, achieving 98.5% accuracy - outperforming ANN, random forest, decision tree, and SVM. Research presented at NUS.

GitHub

Explorer

Android app helping visually impaired users localize and identify objects within a camera frame using real-time image processing and voice feedback.

GitHub

Technical Skills

Programming: Python, Java, C++, SQL, JavaScript

Machine Learning & AI: Deep RL, LLMs, RAG, Agentic AI, CNN / ANN, YOLO, Sim2Real, Scikit-learn

Frameworks & Tools: Spring Boot, REST APIs, MERN Stack, Android Studio, Gradle, Git

Databases: Oracle DB, MySQL, MongoDB

Research Areas: Math Reasoning in LLMs, Sample Efficiency in RL, Human Feedback (RLHF), Big Data

Presentations & Honors

Presentation @ 3rd WDT Semi-Annual Review
University of Maryland Baltimore County May 2025

Presenter @ 2025 COEIT Open House
University of Maryland Baltimore County Mar 2025

Presenter @ PyCon 2024
Python Software Foundation May 2024

Presenter @ PyCon 2023
Python Software Foundation Apr 2023

Presenter @ AI & Autonomy for Multi-Agent Systems (ArtiAmas) Annual Review 2022
University of Maryland College Park May 2022

Presenter @ RetriEVER Empowered Student Success + Research + Community 2022
University of Maryland Baltimore County Apr 2022

1st Place - Human-Guided System Adaptation (HSA) Science Hackathon
Columbia University · DEVCOM Army Research Laboratory Aug 2022

Gold Medal - Python (HackerRank)
HackerRank Apr 2020

Kernel Bronze Medal - Kaggle
Kaggle Jan 2020

Best Paper Award - ICIoT 2019
International Conference on Internet of Things (ICIoT 2019) · SRM University Mar 2019

Top 5 Finalist - Formula Bharat 2018
Formula Bharat · Team Camber Racing Jan 2018

SRM CARE - ELAB Activity
SRM Centre for Applied Research in Education · SRM University Mar 2018

Academic Service

- Reviewer - Workshop on Mathematical Reasoning and AI, MATH-AI @ NeurIPS 2025
- Reviewer - Inductive Biases in Reinforcement Learning Workshop, RLC 2025
- Reviewed 3 papers - Association for Computational Linguistics (ACL 2025)
- Talks & Tutorial Reviewer - PyCon 2025